Improving Cantonese translation in existing multilingual translation models

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Abstract

This is the abstract...

1 Related works

1.1 MADLAD

MADLAD As a base for our experiments, and later as a model to finetune, we use the MADLAD model. This is a general translation model, with 10.7B parameters, trained on web-scraped unlabeled data in 419 languages. The dataset contains both Mandarin and Cantonese data, but the amount of Cantonese data is very limited compared to the amount of Mandarin data (approximately 61 times less). The model architecture is a standard transformer-based sequence-to-sequence model, similar to the architecture used in models such as T5[**?**].

2 Introduction

This project explores improving machine translation tasks working with Cantonese text. Although Cantonese has more than 85 million speakers, it is very underrepresented in machine learning and translation, as large training corpusses are rare as the language is most used in its spoken form. This results in many Cantonese NLP tasks simply being handled by models trained mostly or exclusively on Mandarin language data.

In this project, we explore how both unsupervised and supervised learning can be applied to improve the quality of translation tasks. For this, we use the professionally translated data in the SMOL[1] and GATITOS[3] datasets and the unlabeled data in the dataset collected by Dare et al.[2].

The first experiment performed was a simple token replacement. As there is overlap between Cantonese and Mandarin characters, we replace the tokens that are different in the Cantonese input with their Mandarin equivalents before feeding the text to a pre-trained MADLAD[5]. This simple approach showed some promise, but was limited by not being able to handle differences in grammar and differences in meaning of characters that are the same in both languages.

We use the unlabeled data to train Cantonese token embeddings and then transform these embeddings into the embedding space of a pre-trained English-Mandarin model, MADLAD, using the token-level translations in GATITOS. This TODO: Results of this.

We then finetune MADLAD on the SMOL dataset and evaluate the performance of the finetuned model on a held-out test set from SMOL. TODO: Results of this.

3 Datasets

3.1 SMOL

SMOL[1] (Set of Maximal Leverage) is a high-quality translation corpus from English to more than a hundred low-resource languages. SMOL includes SMOLSent and SMOLDoc, which represent sentence-level and document-level translations is nicely accompanied by Gatitos[3] which provides token-level translations across many of the same languages. In this project specifically, we use the English-Cantonese portion of SMOL and GATITOS, where there are translations on all three levels of data.

3.2 Monolingual Cantonese corpus by M. Dare, et. al

In their work on unsupervised Mandarin-Cantonese machine translation, Dare et. al.[2] collected a large corpus of monolingual Cantonese sentences from various online sources. In total 912258 lines were collected by scraping and cleaning data from Wikipedia, Instagram, and a few other sites including a few corpusses compiled for previous academic work. Common for most sources that form the corpus is that the content is mostly user-generated text, which often includes non-standard spellings and colloquial language, which is typical for Cantonese as a language. We use this corpus to train Cantonese token embeddings for the embedding transform experiment described in Section 3.

4 Baseline

We evaluate our model on a subset of the SMOLSent dataset using BLEURT.

5 Experiments

5.1 Token replacement

Since the MADLAD model appears to have difficulty with sentences that contain words that are native to written Cantonese, we do an experiment where we replace words that are native to written Cantonese with their Mandarin Chinese translations. To find words that are native to written Cantonese we use CantoDict[7], a collaborative project where Cantonese speakers annotate Cantonese characters with information about their meaning, pronounciation and usage in written Cantonese and Mandarin. If a character in a sentence is used in Cantonese but not used in Mandarin, we try to find its translation in the Gatitos dataset and do a token-level replacement as preprocessing to MADLAD inference. This is a very naive approach, as we do not take context into account when replacing tokens, and when a character has multiple possible translations we simply choose the first one listed in Gatitos. This can also be seen in the results, listed in section 6.1.

5.2 Embedding transform

From the promising results of the token replacement experiment 6.1, we believe that giving the model more knowledge of Cantonese characters could significantly improve its Cantonese translation ability.

If we were to simply train a model on Gatitos token-level data it might get slightly better at some characters in the small Gatitos dataset, but we would not be able to generalize the model to become better at a significant amount of Cantonese vocabulary.

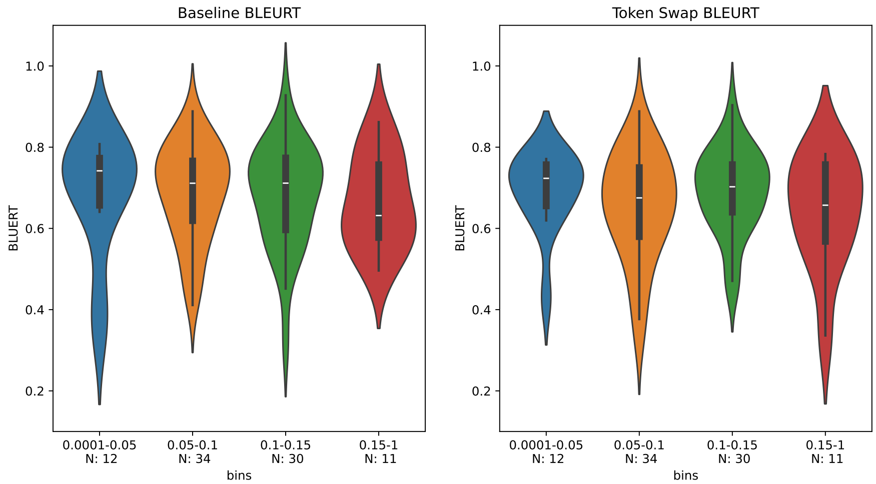

Figure 1: BLEURT scores for token replacement experiment, binned by ratio of Cantonese-only characters in the source sentence.

Instead we propose the following experiment: We train embeddings on a large corpus of unlabeled Cantonese data, hoping to get embeddings that better represent semantic similarity between Cantonese tokens than the embeddings of MADLAD do. We then perform an embedding transform from our Cantonese embeddings to MADLAD's Chinese embedding space using token-level translations from Gatitos. Specifically, we train a transformation matrix W such that for a specific Cantonese-Chinese token embedding pair $E_{canto}, E_{mandarin}$ the embedding transformation $E_{canto}W$ approximates $E_{mandarin}$. This training was simply done optimizing the loss function $||E_{canto}W - E_{mandarin}||$ with the ADAM-algorithm[4].

For training the Cantonese embeddings we use the Monolingual Cantonese corpus by M. Dare, et. al. to train a Word2Vec[6] model.

5.3 SMOL finetuning

We do back-translation finetuning for the MADLAD model on the SMOLSent and SMOLDoc English-Cantonese datasets. We flatten the SMOLDoc dataset such that it consists of sentence pairs like SMOLSent, and combine the two datasets using a 80/10/10 split for training/validation/test.

6 Results

6.1 Token replacement

After performing token replacement and translating the resulting sentences with MADLAD, we calculate BLEURT scores. On average, the token replacement approach performs slightly worse than the baseline, with an average BLEURT score of 0.6704 compared to the baseline's 0.6807. We do however observe that many sentences do see a significant improvement, both in BLEURT score and in translation quality as perceived by humans. At the same time, many sentences also see a significant drop in BLEURT score and translation quality. A few examples of this can be seen in the table **??**. To check whether the ratio of Cantonese characters in the source sentence has an effect on the performance of the token replacement approach, we bin the sentences by their ratio of replaced characters and plot violin plots of the BLEURT scores for each bin in figure 1. We see no real trend in the results, with both the baseline and token replacement approach performing similarly across all bins, although there is a slight tendency for the token replacement approach to perform better on sentences with a higher ratio of Cantonese characters.

Source	Baseline	Token Replacement	Δ BLEURT
Rohan was my sibling who always flew a kite at noon, even if it was cloudy.	The bridge is opened to the public at noon, and closed at midnight.	Rohan is my brother, and he flies kites at noon every day, even when the sky is dark.	0.449354
From 10 feet away you could tell the bottom of the boat was in rough shape.	It was very bad from 10 feet away.	The bottom was in poor condition and visible from a distance of 10 feet;	0.246233
It includes a legend of a devil beast, a century-old family curse.	It includes a legendary demon beast, a family curse that has lasted for hundreds of years.	He is the one who has been the master of the witchcraft for centuries.	-0.379743
Bear in mind, constantly try to purchase silk plants, as they tend to appear like the genuine thing.	Remember, keep trying to buy silk plants because they look like real ones.	They are often used to make the plant's leaves and flowers look like they are actually flowers.	-0.291243

6.2 Embedding transform

6.3 SMOL finetuning

We finetune with a learning rate of 10^{-4} and a batch size of 8 with a weight decay of 0.01 using a A100 GPU on Google Colab. After 1 epoch (784 steps) the model starts to overfit as can be seen on figure **??**, so we use a checkpoint of the model saved after the first epoch.

7 Discussion

7.1 Limitations

In the embedding transform experiment we may have achieved better results if we had pretrained a T5 model on our embeddings using masked sentence modelling, like MADLAD had used.

7.2 Other methods

Non-linear transformations of the embeddings, such as applying ReLU or sigmoid activation functions... TODO: Why we did not try this, too likely to overfit given the size of Gatitos

References

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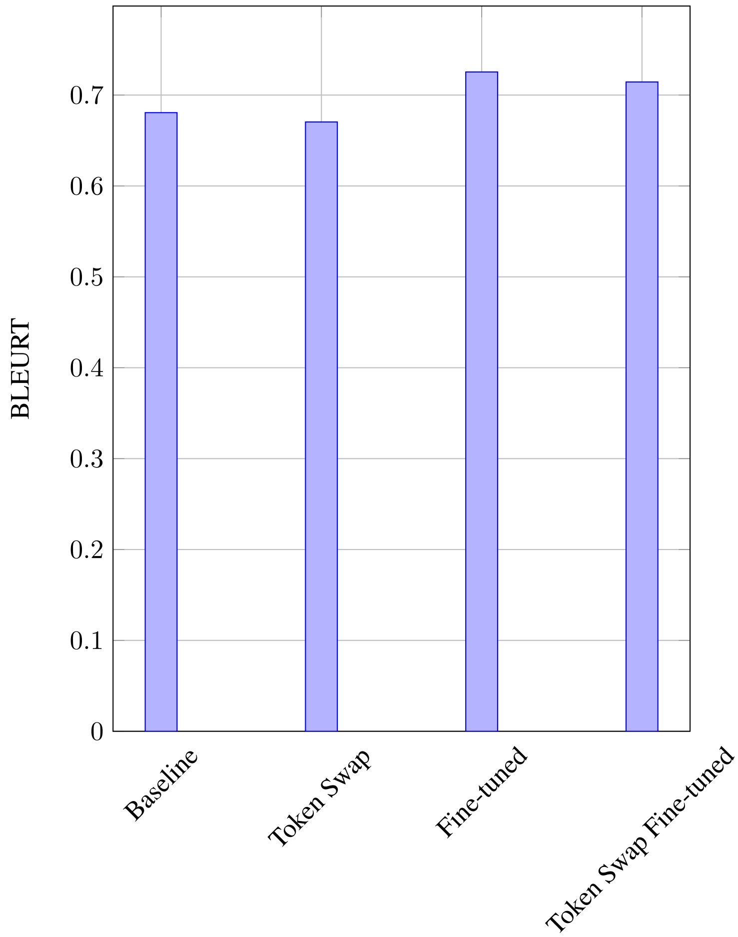

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